Practical 01

Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

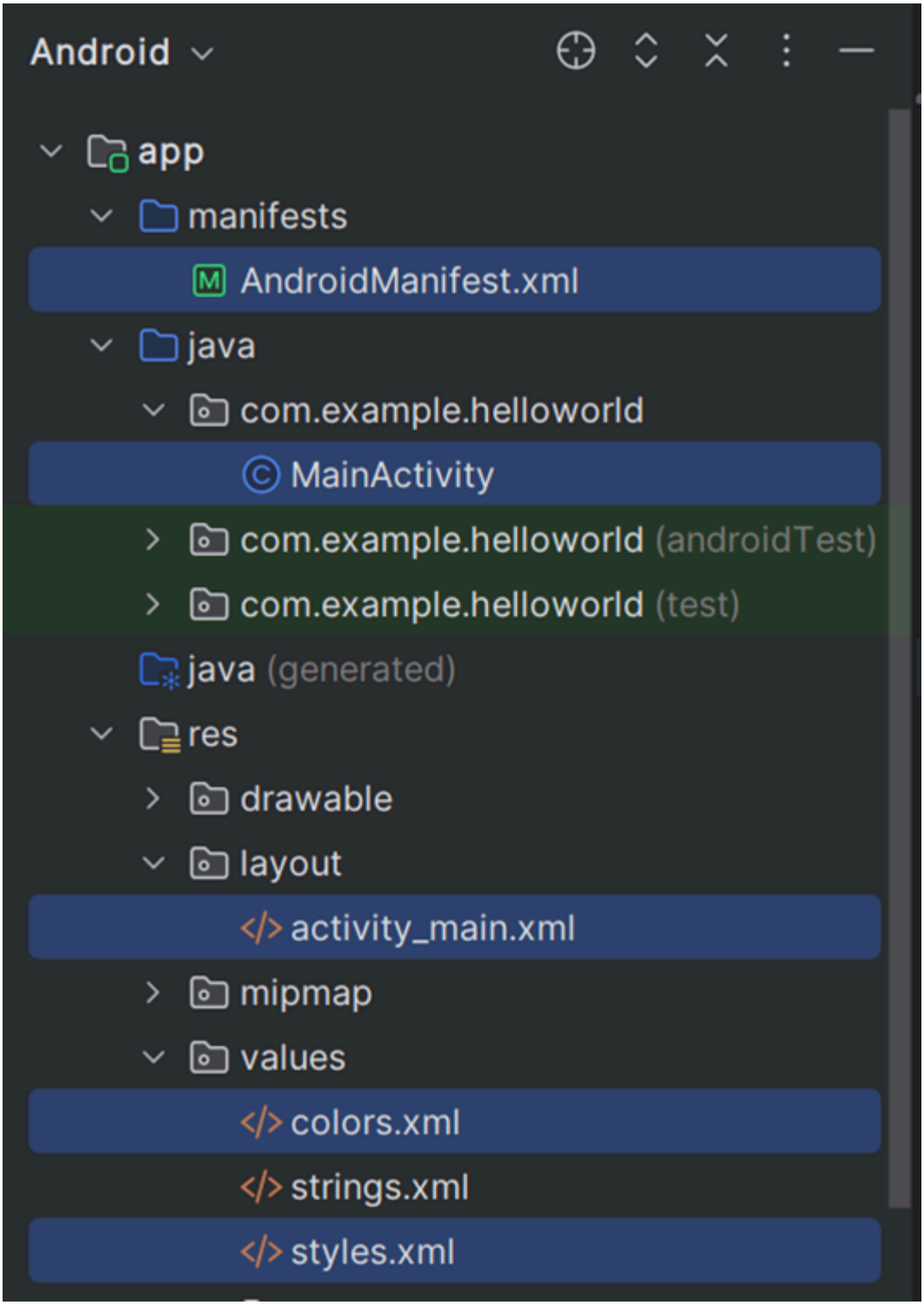
Project 1: Hello World

Project Name: HelloWorld

Template: Empty Views Activity

Project Architecture:

Total 5 files


Android
app
manifests
AndroidManifest.xml
java
com.example.helloworld
MainActivity
com.example.helloworld (androidTest)
com.example.helloworld (test)
java (generated)
res
drawable
layout
activity_main.xml
mipmap
values
colors.xml
strings.xml
styles.xml

I

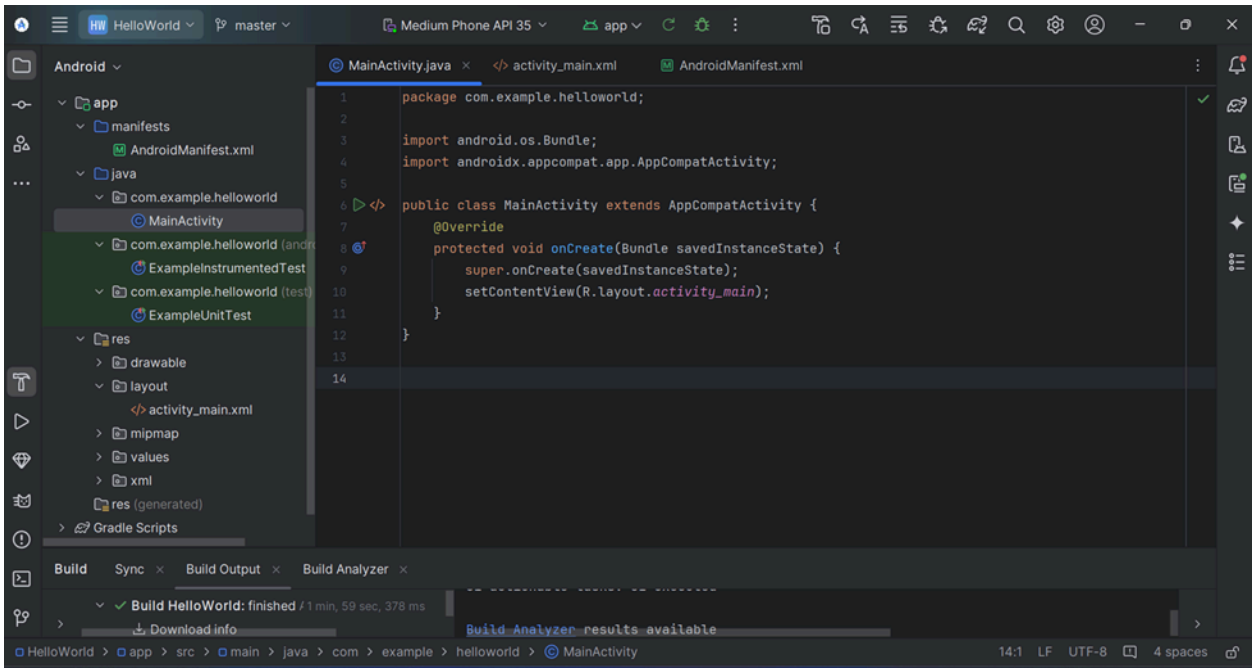
Filename: MainActivity.java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
         super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
}
```

Path: app>java>com.example.helloworld>MainActivity.java



II

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

        xmlns:android="http://schemas.android.com/apk/res/android"

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:orientation="vertical"

        android:gravity="center"

        android:padding="16dp">


        <TextView

        android:id="@+id/text_hello"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello, World!"

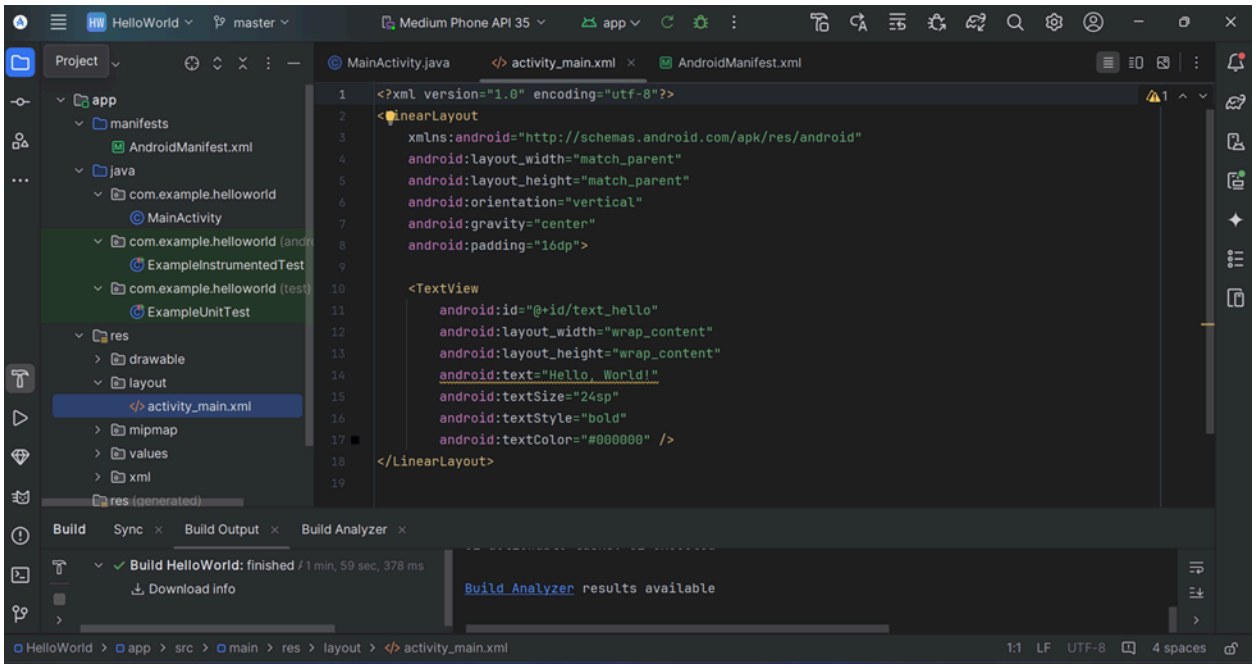
        android:textSize="24sp"

        android:textStyle="bold"

        android:textColor="#000000" />

</LinearLayout>
```

Path: app>res>layout>activity_main.xml



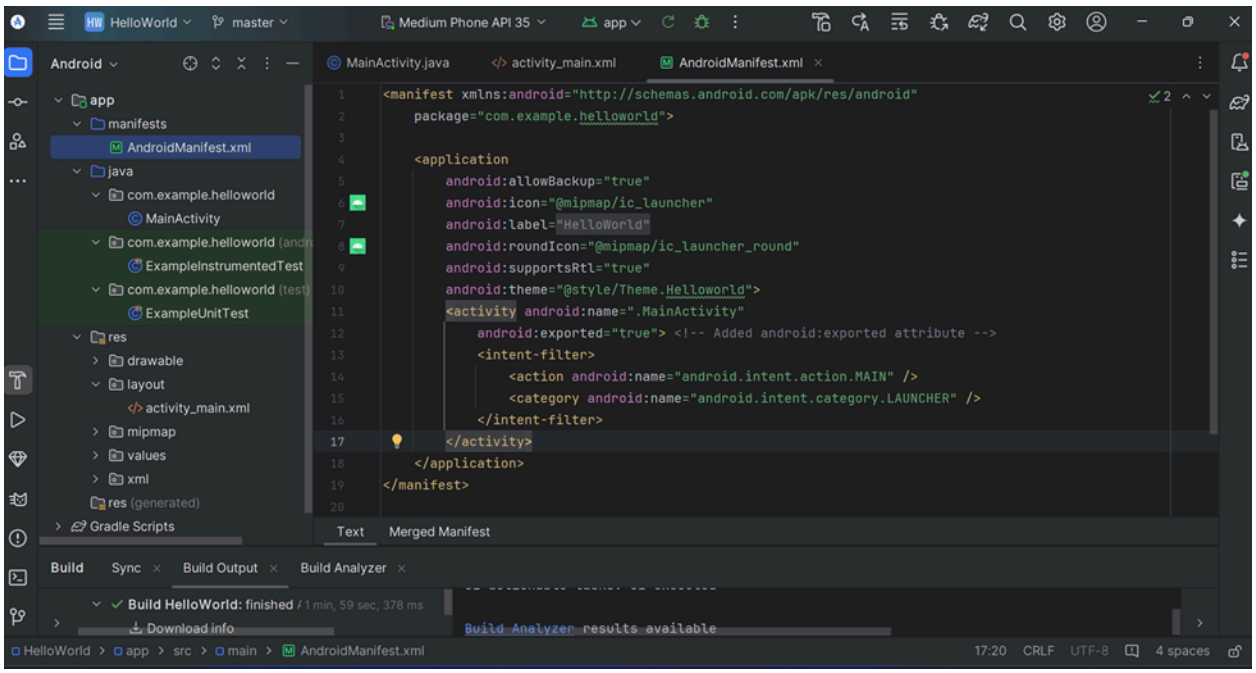
III

Filename: AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
        package="com.example.helloworld">
        <application
    android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Helloworld">
        <activity android:name=".MainActivity"
        android:exported="true"> <!-- Added android:exported attribute -->
        <intent-filter>
                <action android:name="android.intent.action.MAIN" />
          <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
```

```
        </activity>
        </application>
</manifest>
```

Path: app>manifest>AndroidManifest.xml

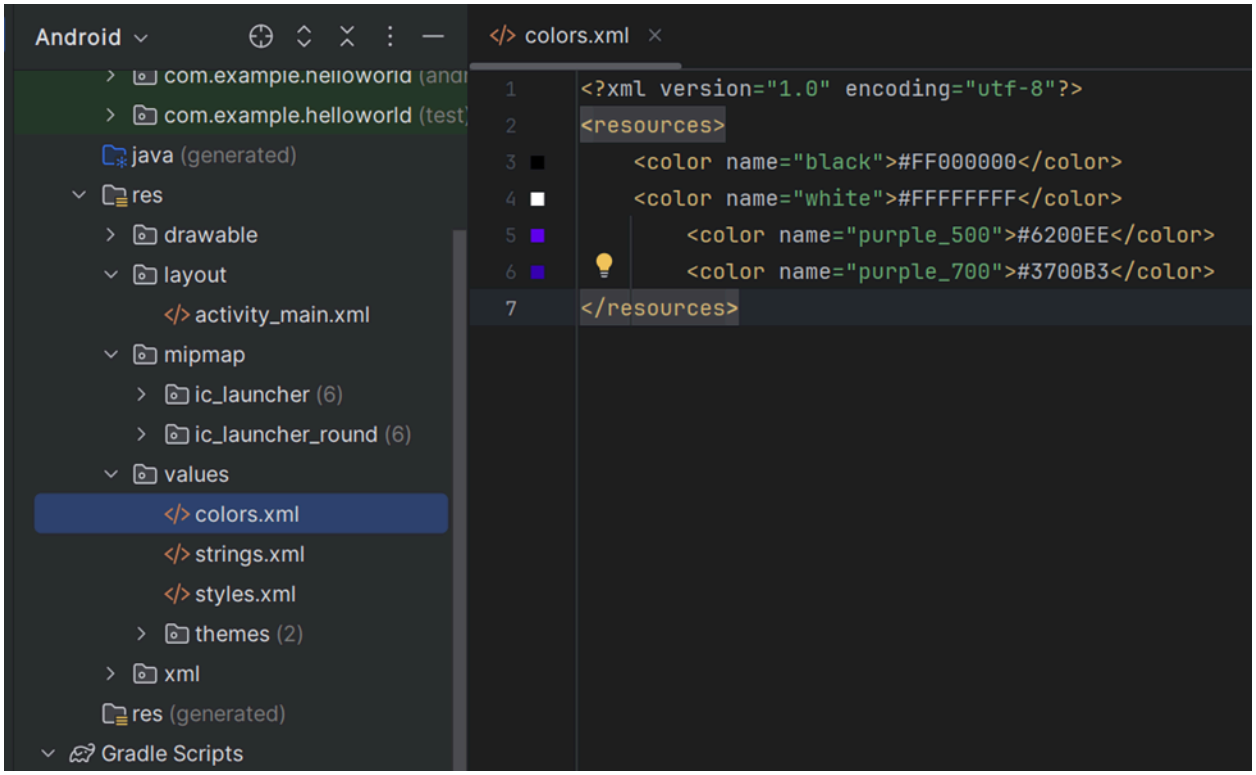


IV

Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="purple_500">#6200EE</color>
      <color name="purple_700">#3700B3</color>
 </resources>
```

Path: app>res>values>colors.xml

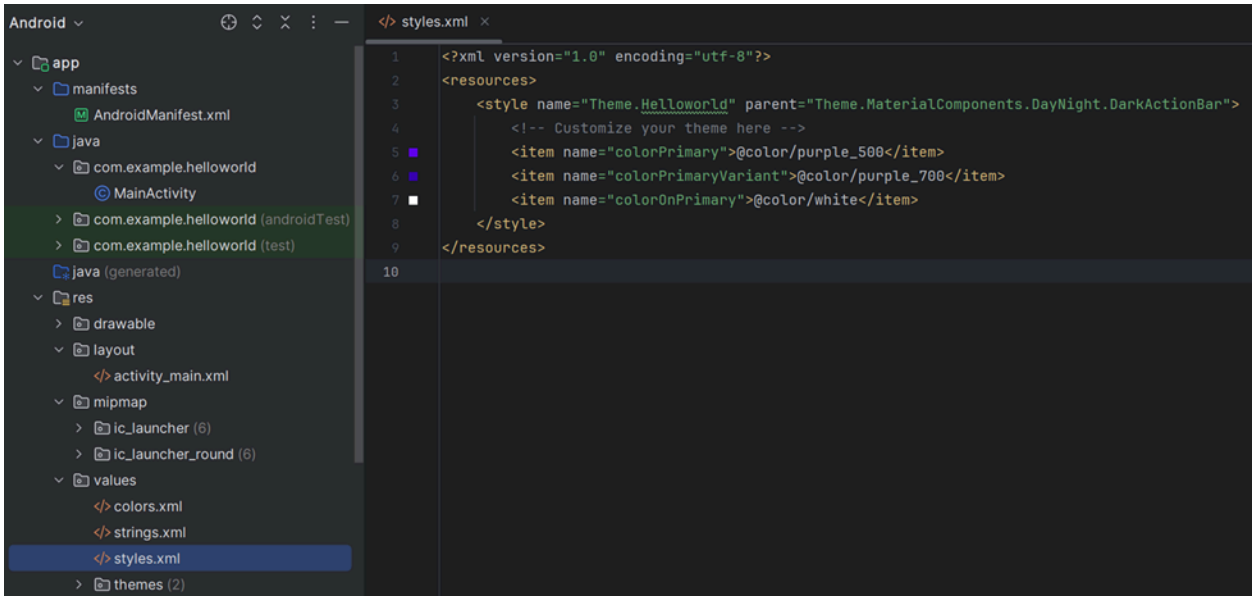


V

Filename: styles.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="Theme.Helloworld"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
      <!-- Customize your theme here -->
      <item name="colorPrimary">@color/purple_500</item>
      <item name="colorPrimaryVariant">@color/purple_700</item>
      <item name="colorOnPrimary">@color/white</item>
      </style>
 </resources>
```

Path: app>res>values>styles.xml



VI (Prebuild Project Code: To ensure proper project detail: No need to update until needed)

Filename: build.gradle.kts (Module:app)

```
plugins {
      alias(libs.plugins.android.application)
 }

 android {
      namespace = "com.example.helloworld"
      compileSdk = 35

      defaultConfig {
      applicationId = "com.example.helloworld"
      minSdk = 24
      targetSdk = 35
      versionCode = 1
      versionName = "1.0"

      testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
      }

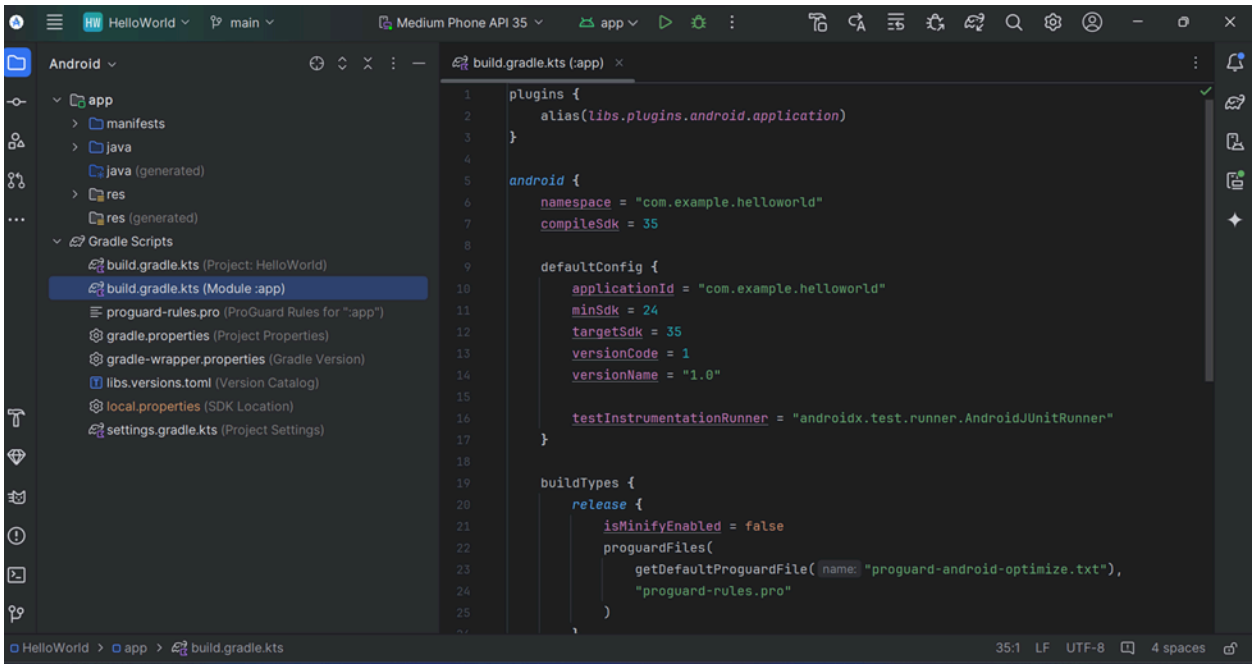
      buildTypes {
      release {
            isMinifyEnabled = false
            proguardFiles(
                  getDefaultProguardFile("proguard-android-optimize.txt"),
                  "proguard-rules.pro"
            )
      }
```

```
        }
        compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
        }
}

dependencies {

        implementation(libs.appcompat)
        implementation(libs.material)
        implementation(libs.activity)
        implementation(libs.constraintlayout)
        testImplementation(libs.junit)
        androidTestImplementation(libs.ext.junit)
        androidTestImplementation(libs.espresso.core)
}
```

Path: Gradle Scripts > build.gradle.kts

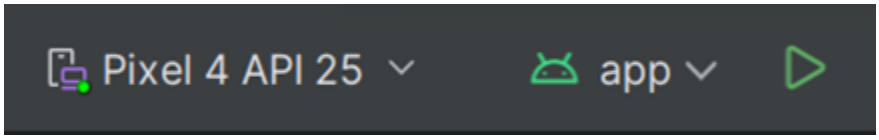


```
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

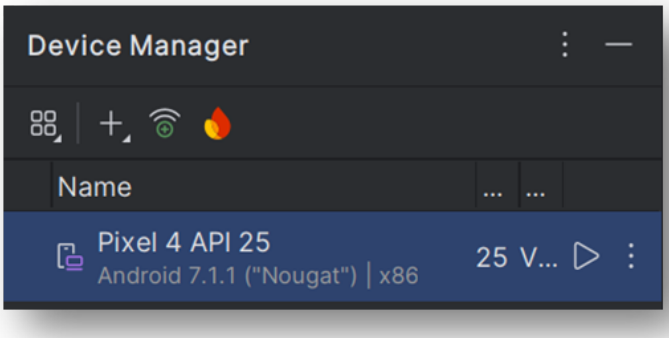
dependencies {

    implementation(libs.appcompat)
    implementation(libs.material)
    implementation(libs.activity)
    implementation(libs.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.ext.junit)
    androidTestImplementation(libs.espresso.core)
}
```

Above the Android Studio near Main Menu Click the Play button to Run the Program



Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat



Output:

